

BTCS301T — COMPUTER VISION

QUIZ 1 — Image Formation & Image Processing

Q1. In the pinhole camera model, an image is formed by light rays passing through:

Answer: A. A single small aperture

Q2. Radiometry, as used in image formation, primarily deals with the measurement of:

Answer: B. Light energy/intensity

Q3. Digitization of an analog image consists of which two processes?

Answer: B. Sampling and quantization

Q4. Spatial resolution of a digital image is primarily determined by:

Answer: B. Sampling rate

Q5. The number of gray levels used to represent pixel intensity is determined by:

Answer: B. Quantization

Q6. Which transformation includes only rotation and translation (no scaling or shearing)?

Answer: B. Rigid transformation

Q7. A key property preserved by an affine transformation is:

Answer: B. Parallelism of lines

Q8. Histogram equalization is primarily used to:

Answer: B. Improve image contrast

Q9. Which of the following kernels is typically used for image blurring?

Answer: C. Averaging/Gaussian kernel

Q10. A high-pass spatial filter primarily enhances:

Answer: B. Edges and fine details

Q11. The Fourier transform converts an image from the:

Answer: B. Spatial domain to frequency domain

Q12. In the frequency domain, sharp edges in an image correspond to:

Answer: B. High-frequency components

Q13. The HSV colour space separates an image into:

Answer: B. Hue, Saturation, Value

Q14. The image negative transform is given by $s = L - 1 - r$, where L is:

Answer: A. The number of gray levels

Q15. Convolution in the spatial domain is equivalent to which operation in the frequency domain?

Answer: C. Multiplication

Q16. Which pixel transform is commonly used for removing salt-and-pepper noise?

Answer: A. Median filter

Q17. An identity convolution kernel applied to an image will:

Answer: C. Leave the image unchanged

Q18. In the thin-lens model, the relationship between object distance, image distance, and focal length is derived from:

Answer: A. Similar triangles / lens equation

Q19. A projective transformation, unlike an affine transformation, does NOT necessarily preserve:

Answer: B. Parallelism of lines

Q20. Removing high-frequency periodic noise from an image is most effectively done using:

Answer: B. Frequency-domain (Fourier) filtering